

Chapter 27

■ Project Scheduling

Slide Set to accompany

Software Engineering: A Practitioner's Approach, 7/e

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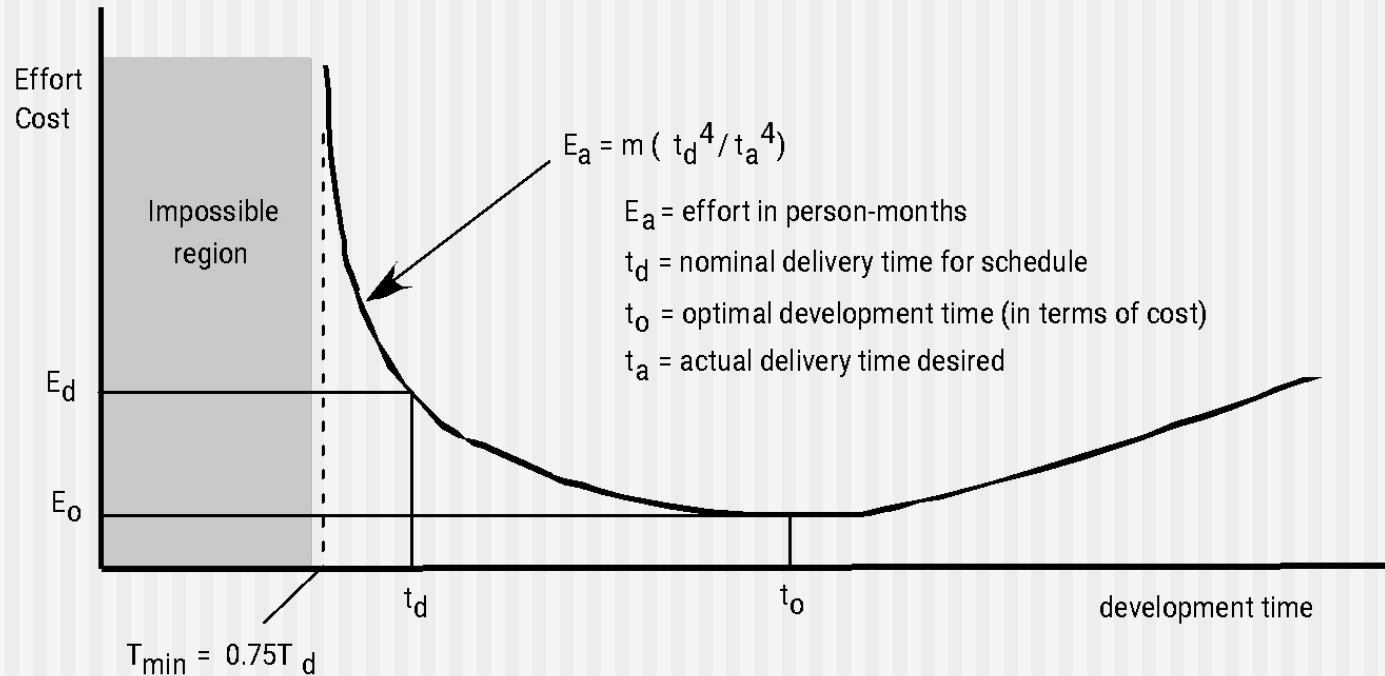
Why Are Projects Late?

- an **unrealistic deadline** established by someone outside the software development group
- **changing customer requirements** that are not reflected in schedule changes;
- an **honest underestimate** of the amount of effort and/or the number of resources that will be required to do the job;
- **predictable and/or unpredictable risks** that were not considered when the project commenced;
- **technical difficulties** that could not have been foreseen in advance;
- **human difficulties** that could not have been foreseen in advance;
- **miscommunication** among project staff that results in delays;
- a failure by project management to recognize that **the project is falling behind schedule** and **a lack of action to correct the problem**

Project Scheduling Principles

- **compartmentalization**—define distinct tasks
- **interdependency**—indicate task interrelationship
- **effort validation**—be sure resources are available
- **defined responsibilities**—people must be assigned
- **defined outcomes**—each task must have an output
- **defined milestones**—review for quality
- Time allocation

Effort and Delivery Time

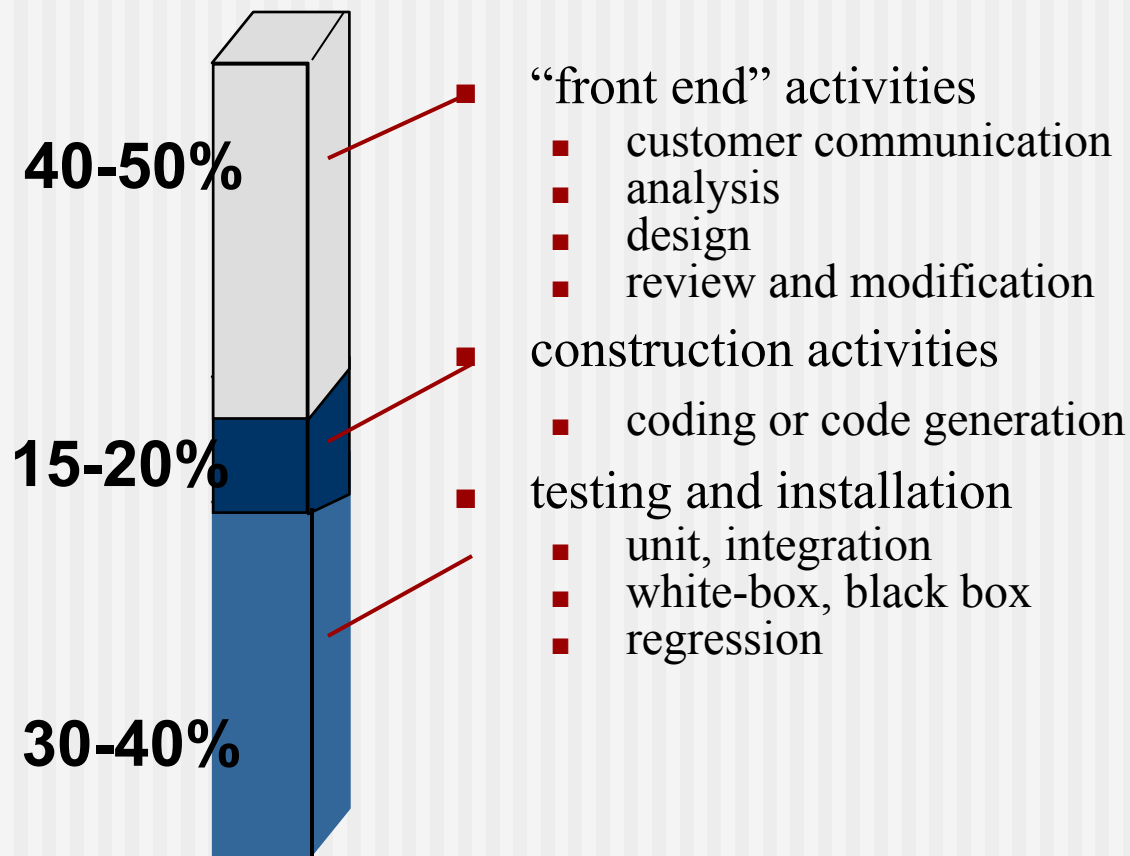


The Relationship Between People and Effort cont..

Curve provides an indication of the relationship between effort applied and delivery time for a software project. A version of the curve, representing project effort as a function of delivery time, is shown in Figure.

The curve indicates a minimum value t_o that indicates the least cost for delivery (i.e., the delivery time that will result in the least effort expended). As we move left of t_o (i.e., as we try to accelerate delivery), the curve rises nonlinearly. Although it is possible to accelerate delivery, the curve rises very sharply to the left of t_d .

Effort Allocation



Defining Task Sets

A task set is a collection of software engineering work tasks, milestones, and deliverables that must be accomplished to complete a particular project.

The task set to be chosen must provide enough discipline to achieve high software quality.

But, at the same time, it must not burden the project team with unnecessary work.

Software Project Types

1. *Concept development projects* that are initiated to explore some new business concept or application of some new technology.
2. *New application development projects* that are undertaken as a consequence of a specific customer request.
3. *Application enhancement projects* that occur when existing software undergoes major modifications to function, performance, or interfaces that are observable by the end user. •
4. *Application maintenance projects* that correct, adapt, or extend existing software in ways that may not be immediately obvious to the end user.
5. *Reengineering projects* that are undertaken with the intent of rebuilding an existing (legacy) system in whole or in part.

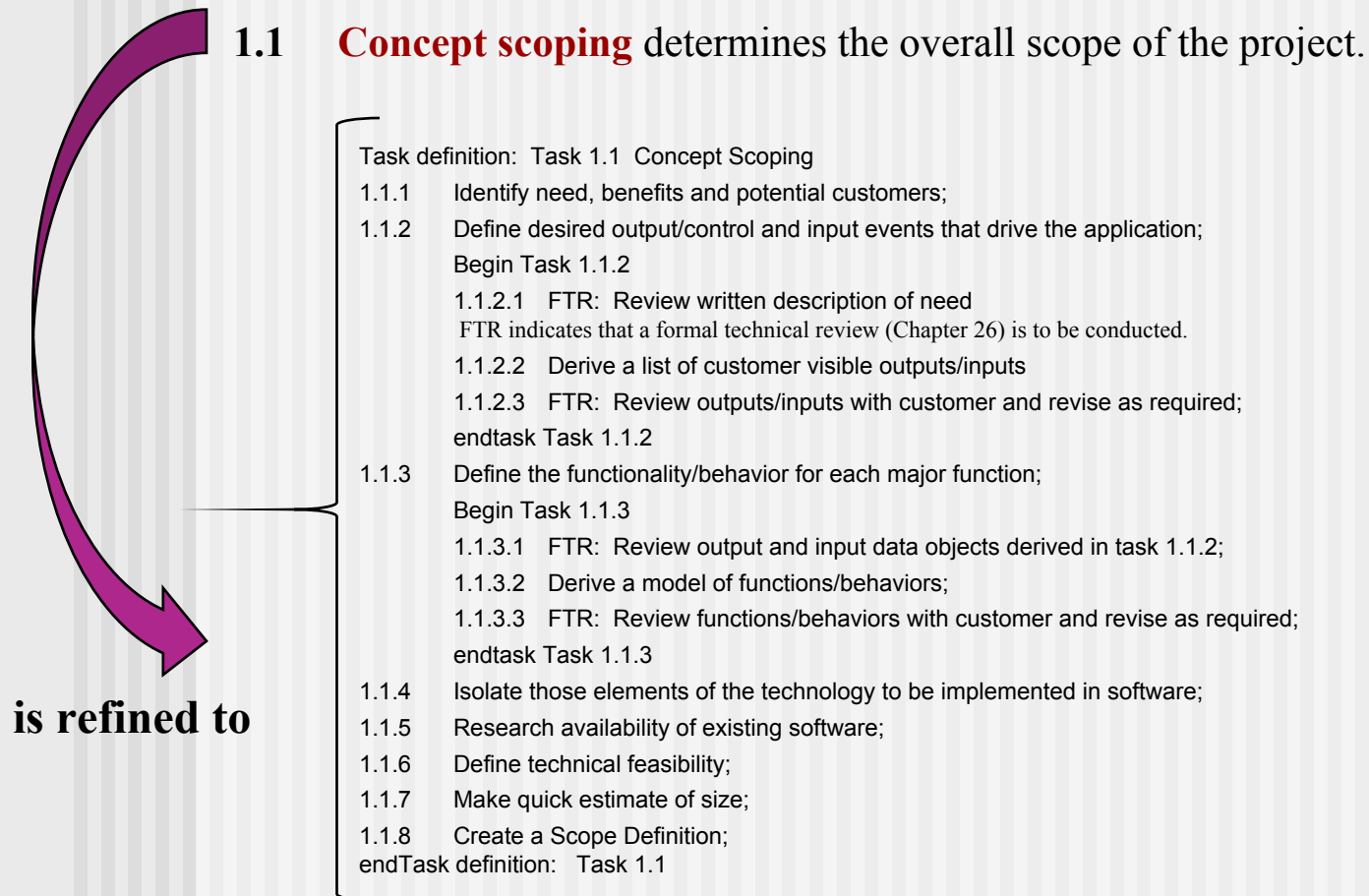
Develop Project Schedule

- determine type of project
- assess the degree of rigor required
- identify adaptation criteria
- select appropriate software engineering tasks

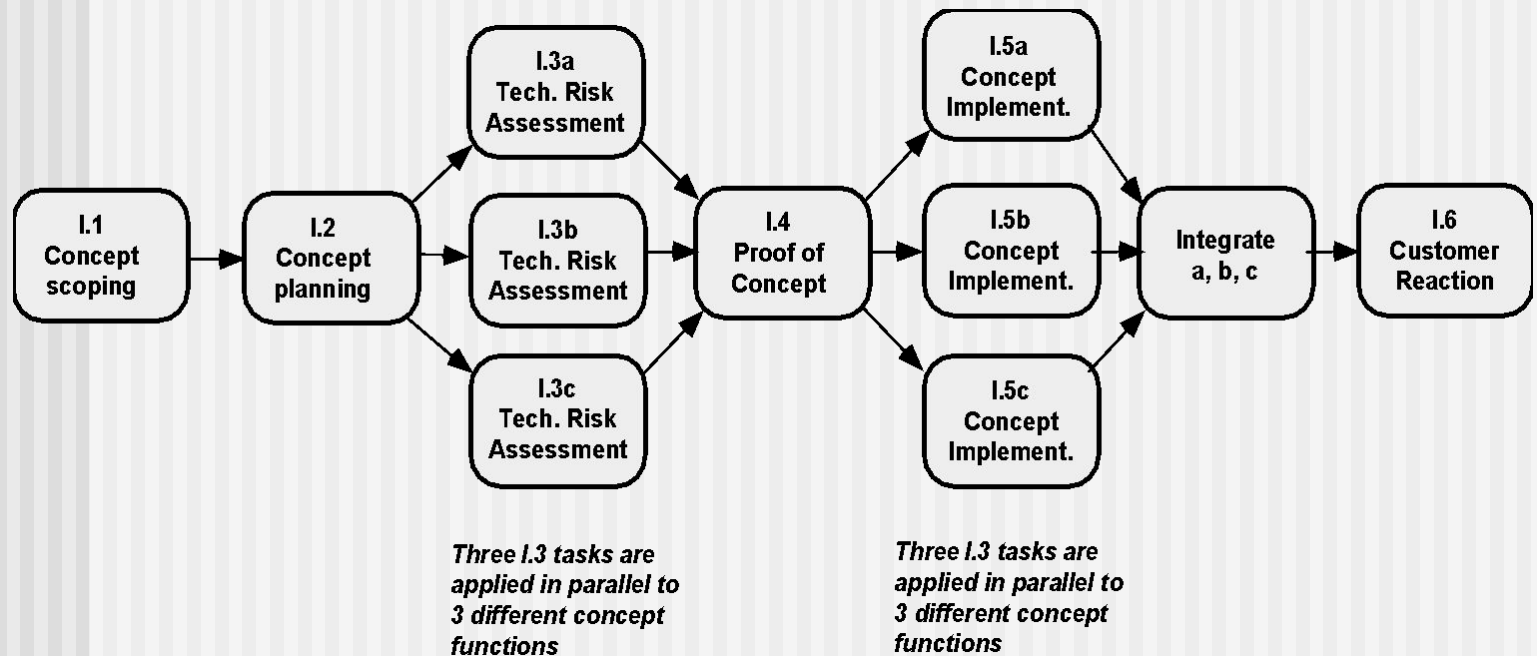
Concept development projects are initiated when the potential for some new technology must be explored. There is no certainty that the technology will be applicable, but a customer (e.g., marketing) believes that potential benefit exists. Concept development projects are approached by applying the following actions:

- 1.1 Concept scoping** determines the overall scope of the project.
- 1.2 Preliminary concept planning** establishes the organization's ability to undertake the work implied by the project scope.
- 1.3 Technology risk assessment** evaluates the risk associated with the technology to be implemented as part of the project scope.
- 1.4 Proof of concept** demonstrates the viability of a new technology in the software context.
- 1.5 Concept implementation** implements the concept representation in a manner that can be reviewed by a customer and is used for "marketing" purposes when a concept must be sold to other customers or management.
- 1.6 Customer reaction** to the concept solicits feedback on a new technology concept and targets specific customer applications.

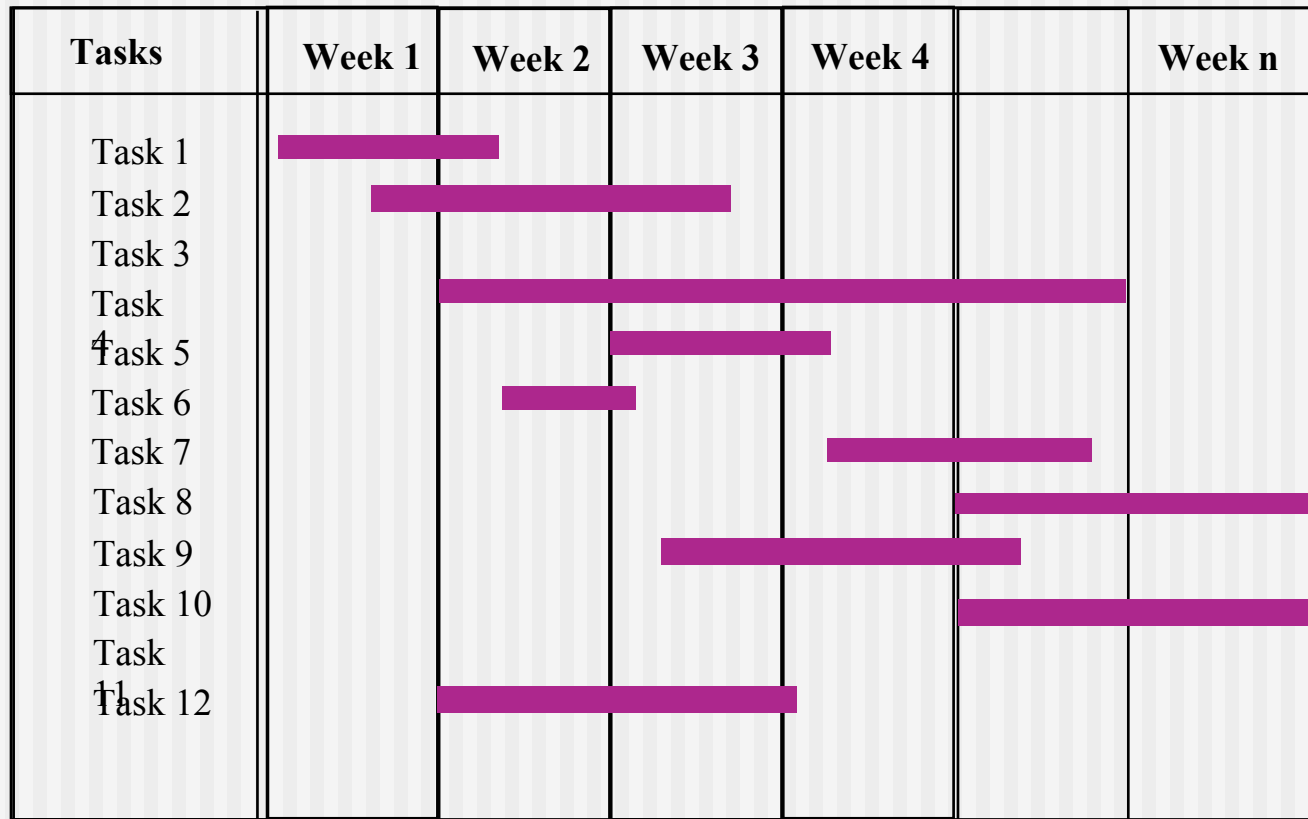
Task Set Refinement



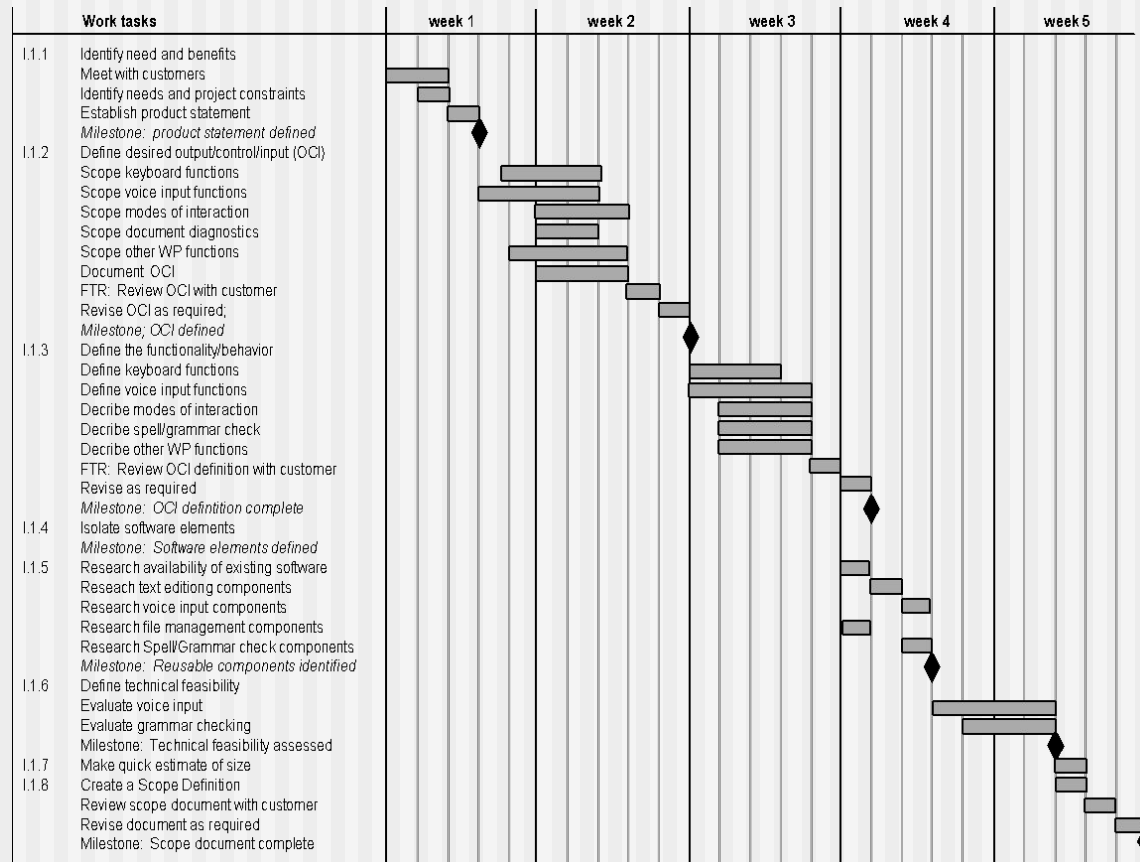
Define a Task Network



Timeline Charts



Use Automated Tools to Derive a Timeline Chart



Schedule Tracking

- conduct periodic project status meetings in which each team member reports progress and problems.
- evaluate the results of all reviews conducted throughout the software engineering process.
- determine whether formal project milestones (the diamonds shown in Figure 27.3) have been accomplished by the scheduled date.
- compare actual start-date to planned start-date for each project task listed in the resource table (Figure 27.4).
- meet informally with practitioners to obtain their subjective assessment of progress to date and problems on the horizon.
- use earned value analysis (Section 27.6) to assess progress quantitatively.

Progress on an OO Project-I

- *Technical milestone: OO analysis completed*
 - All classes and the class hierarchy have been defined and reviewed.
 - Class attributes and operations associated with a class have been defined and reviewed.
 - Class relationships (Chapter 8) have been established and reviewed.
 - A behavioral model (Chapter 8) has been created and reviewed.
 - Reusable classes have been noted.
- *Technical milestone: OO design completed*
 - The set of subsystems (Chapter 9) has been defined and reviewed.
 - Classes are allocated to subsystems and reviewed.
 - Task allocation has been established and reviewed.
 - Responsibilities and collaborations (Chapter 9) have been identified.
 - Attributes and operations have been designed and reviewed.
 - The communication model has been created and reviewed.

Progress on an OO Project-II

- *Technical milestone: OO programming completed*
 - Each new class has been implemented in code from the design model.
 - Extracted classes (from a reuse library) have been implemented.
 - Prototype or increment has been built.
- *Technical milestone: OO testing*
 - The correctness and completeness of OO analysis and design models has been reviewed.
 - A class-responsibility-collaboration network (Chapter 6) has been developed and reviewed.
 - Test cases are designed and class-level tests (Chapter 19) have been conducted for each class.
 - Test cases are designed and cluster testing (Chapter 19) is completed and the classes are integrated.
 - System level tests have been completed.

Earned Value Analysis (EVA)

- Earned value
 - is a measure of progress
 - enables us to assess the “percent of completeness” of a project using quantitative analysis rather than rely on a gut feeling
 - “provides accurate and reliable readings of performance from as early as 15 percent into the project.” [Fle98]

Computing Earned Value-I

- The *budgeted cost of work scheduled (BCWS)* is determined for each work task represented in the schedule.
 - $BCWS_i$ is the effort planned for work task i .
 - To determine progress at a given point along the project schedule, the value of BCWS is the sum of the $BCWS_i$ values for all work tasks that should have been completed by that point in time on the project schedule.
- The BCWS values for all work tasks are summed to derive the *budget at completion, BAC*. Hence,

$$BAC = \sum (BCWS_k) \text{ for all tasks } k$$

Computing Earned Value-II

- Next, the value for *budgeted cost of work performed (BCWP)* is computed.
 - The value for BCWP is the sum of the BCWS values for all work tasks that have actually been completed by a point in time on the project schedule.
- “the distinction between the BCWS and the BCWP is that the former represents the budget of the activities that were planned to be completed and the latter represents the budget of the activities that actually were completed.” [Wil99]
- Given values for BCWS, BAC, and BCWP, important progress indicators can be computed:
 - *Schedule performance index, $SPI = BCWP/BCWS$*
 - *Schedule variance, $SV = BCWP - BCWS$*
 - SPI is an indication of the efficiency with which the project is utilizing scheduled resources.

Computing Earned Value-III

- **Percent scheduled for completion = $BCWS/BAC$**
 - provides an indication of the percentage of work that should have been completed by time t .
- **Percent complete = $BCWP/BAC$**
 - provides a quantitative indication of the percent of completeness of the project at a given point in time, t .
- ***Actual cost of work performed, ACWP***, is the sum of the effort actually expended on work tasks that have been completed by a point in time on the project schedule. It is then possible to compute
 - **Cost performance index, $CPI = BCWP/ACWP$**
 - **Cost variance, $CV = BCWP - ACWP$**